

CODE SPORT



Sandya Mannarswamy

In this month's column, we feature a set of computer science interview questions.

Last month, we had discussed a set of questions on data management and cloud computing. We continue our discussion of computer science questions this month, focusing on machine learning and natural language processing. Let us get started with some tough questions to whet your appetite.

1. In natural language processing, the last couple of years have seen word embedding becoming extremely popular in various NLP tasks such as named entity recognition, sentiment analysis, semantic similarity detection, etc. Most of you would be familiar with Google's word2vec package (<https://code.google.com/p/word2vec/>). What are word embeddings? Can you explain briefly the two major algorithms in word2vec - namely the skip-gram method and continuous bag of words (CBOW) method? Here is a link to an excellent article which provides much of the historical background on word embeddings: <http://gavagai.se/blog/2015/09/30/a-brief-history-of-word-embeddings/>.
2. Over the last few years, we have also seen the rise of deep learning methods in computer vision, pattern recognition, etc. While deep learning achieved considerable popularity with Google's work (<http://www.wired.com/2012/06/google-x-neural-network/>) on using deep learning algorithms to automatically label CAT images, there are still open questions on why deep learning is effective in computer vision, speech recognition and various other pattern-recognition tasks. Can you explain how deep learning is different from traditional methods of machine learning? If you are interested in knowing more about deep learning and how to code on it, I would like to suggest the online tutorial by Andrej Karpathy titled, 'Hacker's Guide to Neural Networks'.
3. Can you explain what a kernel function is, in machine learning? How is it different from a

standard ML classifier, which is based on features?

4. What are 'Support Vector Machines (SVMs)'? Let us assume that you are given a set of data points, some marked red and some blue. You are asked to write a classifier, which can separate the blue and red points. You have written your favourite SVM based classifier for this task. Now you are given Figure 1 as input.



Figure 1: [SVM]

What is the separation boundary your classifier would find for this input? Can you use an SVM based classifier for separation for this input? If yes, explain how. If not, explain why.

5. What is a convex function? What is the significance of convex functions in machine learning algorithms?
6. In supervised machine learning problems, what is the process of cross-validation? How does cross-validation help prevent a model from being over-fitted to training data?
7. In machine learning, can you explain the difference between model parameters and hyper-parameters? Do all models have hyper-parameters? Can you explain what would be the hyper-parameter for the standard linear regression model of the following form:

$$w^T x = y$$

...where x is the feature vector for the data, y is the scalar variable that needs to be predicted and w is the model parameter's matrix?

8. You are running a randomised clinical trial, where you are asked to determine whether a newly discovered drug for cancer is effective. You have a control group of patients who do not get the new